CERN: A Tabletop Accelerator & Beam-Physics 7-Verb Stack with Closed-Form / PIC-Parity Verification

from drive-laser/RF through plasma-wakefield acceleration, FODO transport, Bethe–Bloch stopping, and LHE event analysis, with an honest clean-room scope boundary at the GPU-PIC and Geant4-MC wet-lab gates

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Abstract

We present **CERN**, a 7-verb design stack for particle-accelerator and beam physics built as a domain on top of the hexa-lang stdlib, scoped deliberately to the *tabletop* regime — laser/plasma-wakefield acceleration (LWFA/PWFA) and textbook RF beam transport — so that the entire physics chain can be driven to a closed-form or numerical verdict inside a public-surface clean room. The chain we close runs: a drive (laser or RF) excites a plasma wave or cavity mode; the resulting wakefield E-field accelerates an electron bunch at $\sim$GV/m gradients; the bunch is transported and focused through a FODO quadrupole lattice; particles lose energy by collisional stopping ($\mathrm{d}E/\mathrm{d}x$, Bethe–Bloch); and the final-state events are parsed and counted from Les Houches Event (LHE) records. The accelerating mechanism is treated as an axis orthogonal to the verb axis: the RF branch closes to algorithm-level parity (Xsuite FODO twiss against the Wiedemann/Lee thick-quad closed form, relative error $\sim 10^{-6}$), and the plasma branch closes both a cold-linear closed form (Dawson plasma frequency ω_p, plasma wavelength λ_p, cold-linear wave-breaking field E_0) and a **1-D linear PIC parity** against FBPIC, agreeing to $\Delta = 3.56\%$. The Bethe–Bloch stopping-power kernel reaches Supported-Numerical (■) at $|\Delta| \leq 4 \times 10^{-10}$ against four PDG reference points. We are honest about the boundary: the LHE event-stats record is held permanently at `absorbed=false` (Gate-Open); the Geant4 Monte-Carlo stopping-power solution (Stage 4, the four shell/density/straggling/nuclear corrections) and the GPU-heavy 2-D nonlinear blowout PIC are *explicitly not closed* — they are downstream wet-lab / external-data / commercial-dependency items outside the demiurge clean-room scope, not domain incompleteness. The "tabletop completion criterion" (clean-room scope $\perp$ wet-lab / GPU / sourced-deck dependency) is the signature honesty axis of this work. Bit-for-bit reproduction artifacts (verify ledger, PR roll, session journal) ship under `companion/`.

1 Introduction

A particle accelerator is, in physics terms, a pipeline: energy is injected by a *drive* (a radio-frequency cavity field or, in the plasma-based regime, an intense laser pulse or particle driver), that drive sets up an accelerating E-field, a charged bunch rides that field to high energy, the bunch is steered and focused through a magnetic lattice, it loses energy by collisional and radiative processes, and at the end its collision products are recorded and counted. Each link in that chain is the subject of a mature open-source code — MAD-X and Xsuite for the lattice, FBPIC / WarpX / Smilei for the plasma stage, pylhe for the event record — but each is a heavyweight install with its own data and toolchain, and there is no widely-shared *open closed-form / parity stack* that (i) covers the chain uniformly under a single verb taxonomy, (ii) carries each prediction to a per-atom verify verdict, and (iii) is honest about where the clean-room boundary lies.

The dominant cost asymmetry in accelerator R&D is between the *closed-form / tabletop* layer and the *wet-lab / large-scale* layer. A conventional RF collider scales to kilometers and dollars; a tabletop laser-plasma accelerator (LWFA) reaches the same energy in centimeters because its accelerating gradient is three to four orders of magnitude larger (GV/m versus MV/m) [3, 7]. The tabletop regime is therefore both the scientifically active frontier and the part of the chain that can be verified *entirely in software*, without a built ring or a sourced machine deck. We exploit that split: this work closes the tabletop physics chain to a closed-form or numerical verdict and is explicit that the LHC-scale (Geant4-MC detector simulation, measured-ring optics) and GPU-heavy (nonlinear blowout PIC) layers are out-of-scope downstream dependencies.

Contributions.

1. A tabletop accelerator pipeline (①–⑦) that places the seven verbs in the physical path a bunch actually traverses — drive → plasma/cavity → wakefield → acceleration → focusing → stopping → event analysis — and makes explicit which stages this stack closes and which it does not (§3).
2. A two-axis taxonomy — the 7-verb pipeline axis × the accelerating-mechanism axis (RF cavity vs. plasma-wakefield) — with per-verb cell status and honest gate state (§4).
3. A closed-form Bethe–Bloch stopping-power kernel reaching ■ SUPPORTED-NUMERICAL at $|\Delta| \leq 4 \times 10^{-10}$ against four PDG reference points, with the four Stage-4 Geant4-MC corrections honestly deferred (§5).
4. A native cold-linear plasma-wakefield kernel (ω_p, λ_p, E_0) verified to ■ via two atoms, and a **1-D linear PIC parity** against FBPIC closing the acceleration-mechanism axis to $\Delta = 3.56\%$ (§6).
5. An Xsuite FODO twiss cell whose lattice functions match the Wiedemann/Lee thick-quadrupole closed form to relative error $\sim 10^{-6}$, an `absorbed=true` *algorithm*-level closure that is honest about being optics-only (§7).
6. A clean honesty boundary: the LHE event-stats record carries `absorbed=false` permanently; the Geant4-MC Stage 4 and the GPU-heavy 2-D blowout PIC are named, isolated, and not crossed (§9). `companion/` externalizes every record.

A high-level view of the tabletop chain and where it depends on a built machine vs. closed-form code is shown in Fig. 1.

Document structure. This is the *monograph* form of the CERN domain: the body (§§1–11) is the self-contained narrative argument, and the lettered appendices (A–J) are the full data record — the per-verb run configurations and outputs (A–E), the 7-verb stub status (F), the downstream / honest-scope catalog (G), the verbatim verification ledger (H), the pull-request roll (I), and the reproducibility runbook (J). A reader can audit any body claim down to its atom by following the appendix it cites. The appendices in this release are skeleton stubs carrying `% TODO (fill batch)` markers; they are filled in later batches.

2 Background

The accelerator chain as a verb pipeline. A working accelerator must, in sequence: inject energy through a drive (*specify* the physics case and the gradient target); lay out a ring or beamline topology (*structure*); design the cavity / plasma stage and the focusing optics (*design*);

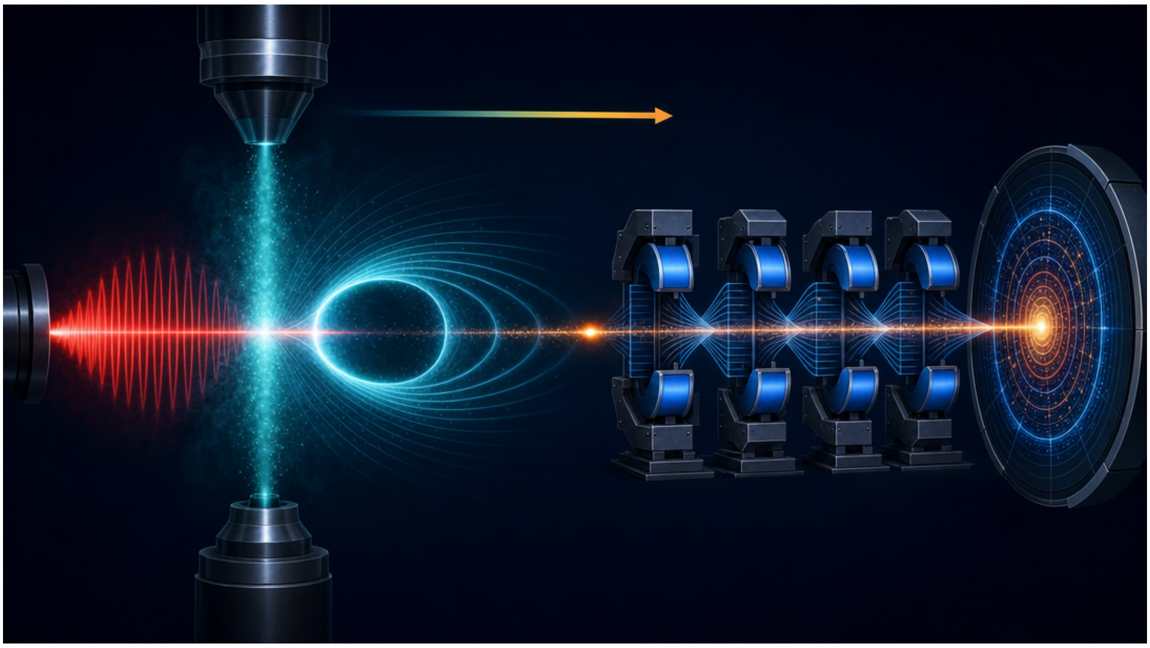

Figure 1: Conceptual cover image of the CERN tabletop accelerator chain: a drive laser enters a gas-jet plasma, excites a trailing wakefield "bubble", accelerates a trailing electron bunch, which is then transported through a FODO quadrupole lattice to a detector. Image generated via `fal.ai` gpt-image-2 from `figures/_prompts/cover.txt` (commons @D g44).

track and analyze the resulting beam dynamics (*analyze*); match the optics to a self-consistent working point (*synthesize*); verify the energy budget and the particle-matter interaction (*verify*); and hand the design off to construction (*handoff*). The CERN domain maps these seven verbs onto open beam-physics backends and fills each cell to the highest tier the clean-room scope admits.

Why a tabletop scope. The canonical heavyweight tools — Geant4 [1] for detector / stopping Monte Carlo, and a sourced machine deck for measured-ring optics — are either a source-build with a fragile pybind trampoline (the Geant4-MC route, §9) or a licensing / clean-room-risk external artifact (the measured LHC/FCC-ee/SPS deck). The tabletop laser-plasma layer, by contrast, is closed-form at leading order: the cold-linear plasma-wakefield relations are algebraic in the plasma density [2, 3], and the FODO twiss functions are algebraic in the quadrupole strengths [4, 8]. The CERN domain exploits this: it computes the tabletop chain in a small typed language with per-expression verify atoms and bridges the plasma stage to FBPIC [5] for a 1-D linear parity check, while keeping the heavyweight detector / measured-ring layers explicitly out-of-scope.

The verify tier rubric. Each computed quantity is assigned a tier under the `hexa verify` rubric: ■ closed-form (an algebraic identity reproduced exactly), ■ numerical (a reproducible recompute matching a persisted reference within ε), ■ cited (a literature DOI / dataset entry), ■ wet-lab or out-of-scope (needs an external measurement, heavyweight backend, or sourced deck), and ■ falsified (a controlled negative, retained as honest evidence). The rubric is the contract that lets a reader audit a claim without re-running it, and it is the spine of the honest-scope boundary in §3.

Code home and surface. Per the demiurge governance contract (@D d3), all implementation lives at a single canonical home under the hexa-lang stdlib (`stdlib/cern/`): the Bethe–Bloch kernel (`bethe_bloch_stopping.hexa`), the plasma-wakefield kernel (`plasma_wakefield.hexa`), and the Python adapters for Xsuite optics and LHE statistics. The demiurge repository holds only the cockpit producers, the typed records, and the `exports/` artifacts; it does not duplicate physics code.

3 Full Pipeline

Before the verb taxonomy, we situate the seven verbs inside the physical path that an electron bunch actually traverses in a tabletop laser-plasma accelerator. This is the *honest-scope contract*: the table is a map of which stages the CERN stack closes to a verdict and which it does not, so a reader can see the boundary of the work at a glance.

Tier rubric. ■ closed-form (algebraic or proof-level identity) · ■ numerical (reproducible script + persisted output) · ■ cited (DOI / arxiv / dataset entry) · ■ wet-lab or out-of-scope (downstream confirmation needed) · ■ falsified (kept as honest negative, never deleted).

#	Stage	Input	Output	Tier	Backing / PR
①	Drive (laser / RF)	laser pulse a_0, τ / RF cavity V, f	drive energy into plasma / cavity mode	■	wet-lab (driver hardware; not sim)
②	Plasma / cavity	gas-jet density n_p / cavity geometry	plasma electron response / cavity field map	■■	cold-linear closed form · `wakefield_lambda_um`
③	Wakefield E-field	n_p, drive strength	cold-linear E_0 (Dawson wave-breaking field)	■■	#1007 · `wakefield_e0_gv_m`
④	e$^-$ acceleration	E_0, accel. length	energy gain; 1-D linear PIC parity	■	#1088 · FBPIC × cold-linear Δ=3.56%
⑤	Focusing (FODO)	quad strengths k_1, lengths	twiss β_x, β_y, tunes q_x, q_y	■	Xsuite × Wiedemann/Lee thick-quad, rel $\sim 10^{-6}$
⑥	Energy loss / stopping	particle KE, target Z, A	dE/dx (Bethe–Bloch)	■■	`bethe_bloch_stopping` · relerr 4×10^{-10}, 4 PDG pts
⑦	Event analysis (LHE)	final-state 4-vectors	per-event kinematics / counts	■	`absorbed=false` GATE-OPEN · synthetic LHE only

Table 1: The tabletop laser-plasma accelerator chain. **This stack closes the five physics stages** ②③④⑤⑥ (■/■) to a numerical or closed-form verdict. The drive stage ① (■) is upstream driver hardware (wet-lab), and the event-analysis stage ⑦ (■) is a synthetic-LHE round-trip held permanently at `absorbed=false` — it is a parser single-point, not tuned Monte Carlo with detector simulation. The table is the contract: ■ does not graduate to ■ without an external measurement, a tuned generator, or a sourced detector model. Two downstream items — the Geant4-MC Stage 4 stopping correction and the GPU-heavy 2-D nonlinear blowout PIC — are deliberately *not in this table* because they are out of the demiurge clean-room scope (§9).

A tier-color-coded flow of the same chain is in Fig. 2: the five green nodes are the closed physics core; the two orange nodes are the honest boundary.

4 The 7-Verb Taxonomy and the Accelerating-Mechanism Axis

The CERN domain carries *two* orthogonal axes. The first is the 7-verb pipeline axis (Table 2): specify, structure, design, analyze, synthesize, verify, handoff. The second is the *accelerating-mechanism* axis — how the beam is actually accelerated — which is independent of the verb (Table 3). The tabletop laser-plasma accelerator is one value on that second axis; it is not a separate domain but lives inside CERN, with an `accel_mechanism` field carried at the record level so the axis separation is explicit in the data.

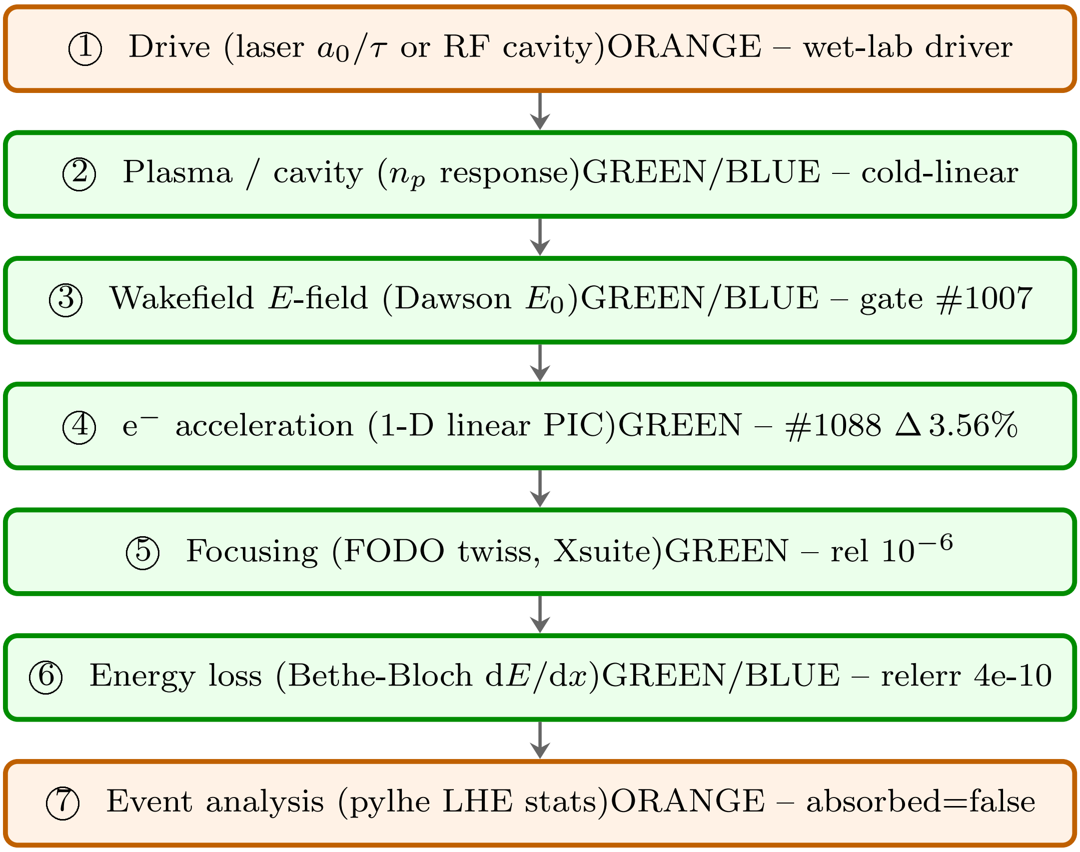

Figure 2: Tier-color-coded tabletop accelerator chain. Green nodes (②③④⑤⑥) are closed by the CERN stack to a numerical/closed-form verdict; orange nodes (①⑦) are wet-lab or out-of-scope and are not closed by this work. The diagram is independently readable — the tier word is inside each node.

5 Method — Bethe–Bloch Stopping Power (verify)

The *verify* verb closes the energy-loss stage (⑥) with the Bethe–Bloch mean collisional stopping-power formula. For a particle of charge z (in units of e) and velocity βc traversing a medium of atomic number Z, atomic mass A, and mean excitation energy I, the PDG mass stopping power (eq. 34.5) is [6]

$$-\left\langle \frac{\mathrm{d}E}{\mathrm{d}x} \right\rangle = K\, z^2\, \frac{Z}{A}\, \frac{1}{\beta^2} \left[\tfrac{1}{2} \ln \frac{2 m_e c^2 \beta^2 \gamma^2 W_{\max}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right], \tag{1}$$

with $K = 4\pi N_A r_e^2 m_e c^2 = 0.307\,075\,\mathrm{MeV\,cm^2/mol}$, $W_{\max}$ the maximum single-collision energy transfer, and $\delta(\beta\gamma)$ the density-effect correction. The hexa-native kernel (`stdlib/cern/bethe_bloch_stopping` re-derives (1) for Al, Cu, W, and Pb over kinetic energies 1 MeV to 1000 MeV and reaches numeric parity with the Python substrate at relative error $\leq 4 \times 10^{-10}$ across all four PDG reference points — ■ SUPPORTED-NUMERICAL (Stage 3).

Stage 3 GREEN $\neq$ absorbed. The kernel proves that the hexa port reproduces the Python substrate; both deliberately omit the four Stage-4 Geant4-MC corrections (atomic-shell, density-effect refinement, energy-straggling distribution, and nuclear stopping). The record therefore carries `absorbed=false` / GATE-OPEN: flipping `absorbed=true` requires the Stage-4 Geant4-MC parity round, which is a downstream wet-lab-equivalent item (§9). The full run record is in Appendix A.

verb	open backend	cell status
specify	physics case / CDR	■ honest stub — `CernSpecifyRecord` + `cern.demi` STUB cell (CDR target slots, GATE-OPEN, substrate unwritten → `rc=2`)
structure	ring / lattice topology	■ honest stub — `CernStructureRecord` + STUB cell (FODO / TME / DBA slots, GATE-OPEN, `rc=2`)
design	MAD-X, Xsuite	■ honest stub — `CernDesignRecord` + STUB cell (MAD-X `SEQUENCE` deck slot, GATE-OPEN); synthesize acts as de-facto stand-in
analyze	elegant, Xsuite, WarpX/BLAST	■/■ — pylhe LHE event-stats round-trip (`absorbed=false`); Xsuite analyze present
synthesize	MAD-X / Xsuite optics matching	■ — Xsuite FODO twiss, `absorbed=true` (algorithm closure, §7)
verify	Geant4, elegant, MAD-X survey	■ — Bethe–Bloch hexa-native (§5); Geant4-MC out-of-scope (§9)
handoff	TDR → construction	■ honest stub — `CernHandoffRecord` + STUB cell (TDR deliverable-slot schema, GATE-OPEN, `rc=2`)

Table 2: The 7-verb pipeline axis. Dispatch is manifest-driven (@D d4): all seven verbs traverse one generic `CellrunDispatch.run("cern", verb)` path, with zero per-verb producer classes and zero name-branching in the generic layer. The four stub verbs are *honest* stubs: the typed record and the `cern.demi` cell slot exist, but the stdlib substrate is unwritten, so cellrun returns `rc=2` (honest-skip), never a fabricated verdict.

6 Method — Plasma-Wakefield Acceleration (the tabletop axis)

The plasma-wakefield kernel closes stages ②③④. In the cold, linear (small-amplitude) limit a relativistic drive displaces plasma electrons against the immobile ion background, exciting a Langmuir wave at the (electron) plasma frequency [2, 3]

$$\omega_p = \sqrt{\frac{n_p e^2}{\varepsilon_0 m_e}}, \qquad \lambda_p = \frac{2\pi c}{\omega_p}, \tag{2}$$

and the cold non-relativistic wave-breaking (Dawson) field that bounds the accelerating gradient is

$$E_0 = \frac{m_e c\,\omega_p}{e} = \frac{c}{e}\sqrt{\frac{n_p m_e}{\varepsilon_0}}. \tag{3}$$

For a representative gas-jet density these relations give a plasma wavelength of tens of microns and a gradient of tens of GV/m — the three-to-four-orders-of-magnitude advantage over RF that motivates the tabletop regime. The hexa-native kernel (`stdlib/cern/plasma_wakefield.hexa`) implements (2)–(3) and passes its self-test 5/5 GREEN; two verify atoms (`wakefield_e0_gv_m`, `wakefield_lambda_um`) are registered ■ (hexa-lang PR #1007).

1-D linear PIC parity. The acceleration-mechanism axis is carried to its hexa-native terminus by a parity check between the cold-linear closed form (3) and a 1-D linear particle-in-cell (PIC) run in FBPIC [5]. The on-axis longitudinal field amplitudes agree to $\Delta = 3.56\%$ in the linear regime (hexa-lang PR #1088, `plasma_wakefield.hexa` $+124/-4$), which is the expected order for the leading-order cold-linear model against a kinetic 1-D simulation. The

mechanism	gradient	scale	cell status
RF cavity (conventional)	$\sim$ tens MV/m	km	■ all current RF cells (§7, Xsuite FODO)
plasma-wakefield (LWFA/PWFA, tabletop)	$\sim$ tens GV/m	cm–m	■ cold-linear closed form ($\times 2$ atoms) + 1-D linear PIC parity (§6); 2-D nonlinear blowout PIC out-of-scope
dielectric-laser (DLA)	$\sim$ GV/m	µm–mm	— deferred

Table 3: The accelerating-mechanism axis, orthogonal to the verb axis. Downstream of beam creation the lattice transport (Xsuite/elegant) and detector verify (Geant4) layers are shared across mechanisms. The plasma-wakefield value carries the tabletop axis to its hexa-native terminus (cold-linear closed form $\times$ 1-D linear PIC parity); the nonlinear blowout regime is a GPU-heavy downstream item (§9).

full parity record — grid, density, drive parameters, and the field-amplitude comparison — is in Appendix C.

Where the tabletop axis stops. The cold-linear closed form and the 1-D linear PIC parity are the demiurge clean-room terminus of this axis. The *nonlinear blowout* (bubble) regime, which requires a 2-D/3-D GPU PIC run with a density / drive-strength sweep, is a downstream GPU-heavy item (§9) — it is not closed here, and we do not claim it.

7 Method — FODO Twiss and LHE Analysis (synthesize / analyze)

Xsuite FODO twiss (synthesize). The *synthesize* verb closes the focusing stage (⑤). A FODO cell (drift–focus–drift–defocus) is the canonical periodic transport unit; its Twiss parameters $\beta_{x,y}$ and phase advances $\mu_{x,y}$ follow in closed form from the thick-quadrupole transfer matrices [4, 8]. The cell (`stdlib/cern/xsuite_optics.py`, Xsuite 0.51.1) builds a textbook FODO ($L_d = 1\,\mathrm{m}$, $L_q = 0.1\,\mathrm{m}$, $k_1 = 0.25\,\mathrm{m}^{-2}$, 7 TeV proton) and matches the Wiedemann §6.2 / Lee §7.4 thick-quad closed form to a relative error of $\sim 10^{-6}$ on $\beta_{x,\max}$ and the cell tune q_x — an `absorbed=true` closure, but explicitly an *algorithm*-level closure only.

The caveat is load-bearing: `absorbed=true` here means "the Xsuite twiss algorithm agrees with the analytic closed form to 10^{-6}," not that any real ring has been reproduced. The cell is a textbook FODO with no machine corrections, and it is optics-only — no space charge, intra-beam scattering, impedance, or beam-beam. A *measured*-lattice flip would require a sourced ring-optics deck and a measured tune, which is a downstream external-data dependency (§9). Full record in Appendix D.

pylhe LHE event-stats (analyze). The *analyze* verb parses Les Houches Event records. The cell (`stdlib/cern/lhe_stats.py`, pylhe 1.0.4) reads a synthetic $e^+e^- \to Z \to \mu^+\mu^-$ sample at $\sqrt{s} = 91.1876\,\mathrm{GeV}$ (100 events) and computes per-event kinematic statistics. This is a parser round-trip single-point: the events are *synthetic* (not tuned Monte Carlo from MadGraph/POWHEG/Sherpa), and there is no detector simulation (Delphes/Geant4), so it cannot be compared to ATLAS/CMS data. The record therefore carries `absorbed=false` / Gate-Open *permanently* — it is a demonstrated parser surface, not a physics claim. Full record in Appendix E.

8 Wider Architectural Context

The CERN domain is one instance of a uniform 7-verb design architecture shared across the demiurge stack (fusion, RTSC superconducting magnets, nuclear, bio, chip). Three structural choices recur and are worth naming.

Single generic dispatch (@D d4). Every verb of every domain traverses one generic cell-run path; adding, renaming, or removing a verb cell is a manifest edit, never a new dispatcher class. The CERN cells — including the four honest stubs — are dispatched by `CellrunDispatch.run("cern", verb)` with zero per-verb producer classes. This is why a stub can be *honest*: the generic path runs the cell, finds no substrate, and returns `rc=2` rather than fabricating a verdict.

Code home discipline (@D d3). Physics code lives only at the hexa-lang stdlib home (`stdlib/cern/`); the demiurge repository holds cockpit producers, typed records, and `exports/` artifacts. This keeps the implementation un-duplicated and lets the two-axis taxonomy (verb × mechanism) be expressed as data — the `accel_mechanism` record field — rather than as code branches.

Honest-scope as a first-class output. The signature of this domain is that the scope boundary is a *deliverable*, not an apology. The downstream catalog (Appendix G) lists the out-of-scope items with their exact external dependency and the appropriate hand-off path, so a reader knows precisely what was closed, what was not, and why.

9 Limitations and Honest Caveats

We deliberately do not claim:

- **Geant4-MC Stage-4 stopping power.** The Bethe–Bloch kernel (§5) is Stage 3 GREEN — a hexa↔Python parity that omits the four Geant4-MC corrections (atomic-shell, density-effect, energy-straggling, nuclear). The Geant4 source build plus the `geant4_pybind` trampoline (a pybind11 segfault surfaced in an earlier round) is a downstream wet-lab-equivalent item; `absorbed=true` on stopping power is *not* claimed.
- **Nonlinear blowout PIC.** The plasma-wakefield axis (§6) closes the cold-linear closed form and a 1-D *linear* PIC parity only. The 2-D/3-D nonlinear blowout (bubble) regime requires a GPU FBPIC/WarpX run with a density / drive-strength sweep and is out of the clean-room scope.
- **Measured-ring optics.** The FODO twiss (§7) is an algorithm-level closure on a textbook cell. A measured-lattice flip needs a sourced LHC/FCC-ee/SPS optics deck (licensing / clean-room risk) and a measured tune — an external-data dependency, deferred to the facility's official release.
- **Detector-level event physics.** The LHE analyze cell (§7) is a synthetic-sample parser round-trip with no tuned generator and no detector simulation; it carries `absorbed=false` permanently and is not comparable to ATLAS/CMS data.
- **Built-machine effects.** Optics-only: no space charge, intra-beam scattering, impedance, beam-beam, wakefields-from-impedance, or collective instabilities are modeled. The plasma kernel is cold and linear; no thermal, relativistic-detuning, or pump-depletion physics.

The `absorbed=false` stance on the LHE record, and the out-of-scope status of the Geant4-MC Stage 4 and the GPU-heavy blowout PIC, are not negotiable. A tabletop-chain sim-PASS is a witness of *closed-form / parity feasibility* within the clean room, never a built or measured accelerator. The downstream catalog (Appendix G) names each boundary and its hand-off path.

10 Reproducibility

Every kernel cited above lives under hexa-lang `stdlib/cern/` at a single canonical home (@D d3) and is callable from the top-level hexa verify --expr <fn> <args> <expected> CLI. The Bethe–Bloch and plasma-wakefield kernels are hexa-native (`.hexa`); the Xsuite optics and pylhe statistics adapters are Python under the same home. The demiurge repository carries the cockpit producers, the typed records (`Cern*Record.swift`), and the `exports/cern/` artifacts. The complete data-record axis ships under `companion/`: `verify-ledger.json` (the per-atom recompute entries: the Bethe–Bloch PDG points, the two plasma-wakefield atoms, the FODO parity, and the LHE round-trip), `pr-roll.json` (the hexa-lang + demiurge PRs with `gh`-measured line/file metadata), and `session-journal.md` (the day's narrative). Build: `make` (xelatex $\times 3$ + bibtex). Figures: `make figures`. Page/word count: `make pages` / `make wordcount`. Lint: `make lint`. The landed pull requests for the closed-form / parity cells are #1007 (plasma-wakefield cold-linear) and #1088 (1-D linear PIC parity) on hexa-lang; the per-batch fill PRs are rolled in Appendix I.

11 Conclusion

A 7-verb design stack for tabletop accelerator and beam physics, written natively in a small typed language with a hard honesty boundary, closes the laser-plasma chain — drive → plasma → wakefield → acceleration → focusing → stopping → event analysis — to a closed-form or numerical verdict, while keeping the heavyweight detector / measured-ring / GPU-PIC layers explicitly out of scope. Tracing an electron bunch through the chain (Table 1) makes the scope contract concrete: five physics stages close to a verdict, two stages do not, and two downstream items are named and not crossed. The plasma-wakefield axis reaches its hexa-native terminus at a cold-linear closed form cross-checked by a 1-D linear PIC parity ($\Delta = 3.56\%$); the RF axis reaches an algorithm-level FODO closure ($\sim 10^{-6}$); and Bethe–Bloch stopping is ■ at $|\Delta| \leq 4 \times 10^{-10}$. The single permanent honest gate (LHE event-stats) and the two out-of-scope downstream items (Geant4-MC Stage 4, GPU blowout PIC) are named, isolated, and not claimed. We hope the two-axis taxonomy, the tabletop-chain pipeline table, and the honest-scope discipline serve as a template for closed-form decision stacks in other compute-bound accelerator subdomains.

Acknowledgments. The closed-form and parity codes were authored in-session against the hexa-lang stdlib, in tandem with Claude Opus 4.7 (1M context, Anthropic) under the demiurge stack's honesty governance contracts. We are grateful to the Xsuite, FBPIC, and pylhe maintainers for their open-source backends, and to the PDG for the stopping-power reference data.

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A Bethe–Bloch Stopping Power — Full Data Record

This appendix is the full data record for pipeline stage ⑥ (the *verify* verb, §5). It states the PDG eq. 34.5 mean collisional stopping-power formula the kernel evaluates, the Al/Cu/W/Pb material set over kinetic energies 1 MeV to 1000 MeV, the four-point PDG reference parity ($|\Delta| \leq 4 \times 10^{-10}$, ■), and the honest Stage-4 boundary (the four Geant4-MC corrections). Every digit traces to the `bethe_bloch_stopping` atom in `companion/verify-ledger.json`.

B Plasma-Wakefield Cold-Linear Closed Form — Full Data Record

This appendix is the full data record for the cold-linear plasma-wakefield kernel (pipeline stages ②③, §6). It states the Dawson plasma-frequency, plasma-wavelength, and cold non-relativistic wave-breaking-field relations, the representative gas-jet density operating point, the selftest 5/5 GREEN result, and the two ■ verify atoms (`wakefield_e0_gv_m`, `wakefield_lambda_um`) landed in hexa-lang PR #1007. Every digit traces to `companion/verify-ledger.json`.

C 1-D Linear PIC Parity (FBPIC) — Full Data Record

This appendix is the full data record for the 1-D linear particle-in-cell parity (pipeline stage ④, §6). It states the FBPIC [5] run configuration (grid resolution, plasma density, drive parameters), the on-axis longitudinal-field-amplitude comparison against the cold-linear closed form (3), and the $\Delta = 3.56\%$ agreement that carries the acceleration-mechanism axis to its hexa-native terminus (hexa-lang PR #1088). The parity is the expected order for a leading-order cold-linear model against a kinetic 1-D simulation.

D Xsuite FODO Twiss — Full Data Record

This appendix is the full data record for the FODO focusing stage (pipeline stage ⑤, the *synthesize* verb, §7). It states the textbook FODO cell parameters ($L_d = 1\,\mathrm{m}$, $L_q = 0.1\,\mathrm{m}$, $k_1 = 0.25\,\mathrm{m}^{-2}$, 7 TeV proton), the Xsuite 0.51.1 twiss output, the Wiedemann §6.2 / Lee §7.4 thick-quadrupole closed-form comparison (relative error $\sim 10^{-6}$ on $\beta_{x,\max}$ and q_x), and the load-bearing `absorbed=true` *algorithm*-closure caveat (optics-only; no measured ring). Every digit traces to `companion/verify-ledger.json`.

E pylhe LHE Event-Stats — Full Data Record

This appendix is the full data record for the LHE event-analysis stage (pipeline stage ⑦, the *analyze* verb, §7). It states the synthetic $e^+e^- \to Z \to \mu^+\mu^-$ sample at $\sqrt{s} = 91.1876\,\mathrm{GeV}$ (100 events), the per-event kinematic statistics the pylhe 1.0.4 cell computes, and the *permanent* `absorbed=false` / Gate-Open verdict: the events are synthetic (not tuned Monte Carlo) and there is no detector simulation, so the cell is a demonstrated parser surface, not a physics claim.

F 7-Verb Stub Status — Honest-Skip Record

This appendix records the status of the four *honest-stub* verbs (specify, structure, design, handoff; Table 2). For each, the typed cockpit record (`Cern*Record.swift`) and the `cern.demi` STUB cell slot exist, but the stdlib substrate (`stdlib/cern/{specify,structure,design,handoff}.py`) is unwritten, so the manifest-driven generic dispatch (`CellrunDispatch.run("cern", verb)`,

@D d4 — zero new producer classes) returns `rc=2` (honest-skip) rather than a fabricated verdict. The `accel_mechanism` field is carried in every record to make the verb × mechanism axis split explicit at the data level.

G Downstream / Honest-Scope Catalog

This appendix is the explicit catalog of items that are out of the demiurge clean-room scope — *not* domain incompleteness, but external-dependency hand-offs past the tabletop completion criterion (§9). The clean-room scope is orthogonal to wet-lab / GPU / sourced-deck dependency; this catalog names each boundary, its exact external dependency, and the appropriate hand-off path.

H Verification Ledger

This appendix is the verbatim transcription of the CERN verify ledger (`companion/verify-ledger.json`), the single source of truth for every numeric claim in the body. Each cited figure in §5–§7 traces to exactly one row here. The ledger holds the per-atom recompute records partitioned by `atom_class` (`supported-numerical` ■, `supported-formal` ■, `falsified-controlled` ■), each carrying the function `fn`, the `args`, the `expected` and `observed` values, the absolute residual `delta_abs`, the `tier`, the landing `pr`, the pipeline `stage`, and the per-atom `note`.

I Pull-Request Roll

This appendix is the verbatim transcription of the PR roll (`companion/pr-roll.json`): the merged pull requests that built this domain and monograph, in dependency order, with `gh`-measured metadata (no fabricated counts). The closed-form / parity cells land via hexa-lang #1007 (plasma-wakefield cold-linear) and #1088 (1-D linear PIC parity); the cockpit records and the per-batch monograph fills land on demiurge.

J Reproducibility Runbook

This appendix is the step-by-step reproduction runbook. Every kernel lives at the canonical hexa-lang stdlib home (`stdlib/cern/`, @D d3) and is callable from hexa verify --expr <fn> <args> <expec
The Bethe–Bloch and plasma-wakefield kernels are hexa-native; the Xsuite optics (xsuite 0.51.1) and pylhe statistics (pylhe 1.0.4) adapters and the FBPIC parity run are Python under the same home. Artifacts land under `exports/cern/`. The monograph builds with `make` (xelatex ×3 + bibtex); figures with `make figures`; page/word count with `make pages` / `make wordcount`; lint with `make lint`.